Day 02 Report: Customer Segmentation

Oct 29, 2026 · @Hiru

Summary

K-Means split 200 mall customers into 5 clear groups by income and spending; the Elbow Method confirmed 5 as the best number.

- The biggest group is average customers (81 people).
- The most valuable group is high income, high spending (39 people, average age 33).
- A high-income group spends very little (35 people), the mall's biggest opportunity.

Data

The Mall Customers dataset has 200 customers, 5 columns and no missing values. There is no answer column, so this is unsupervised learning: the model finds the groups by itself.

Column	Meaning	Used?
CustomerID	Customer number	No
Gender	Male / Female	No
Age	Age in years	For describing groups
Annual Income (k$)	Yearly income in thousands of dollars	Yes
Spending Score (1-100)	How much they spend at the mall	Yes

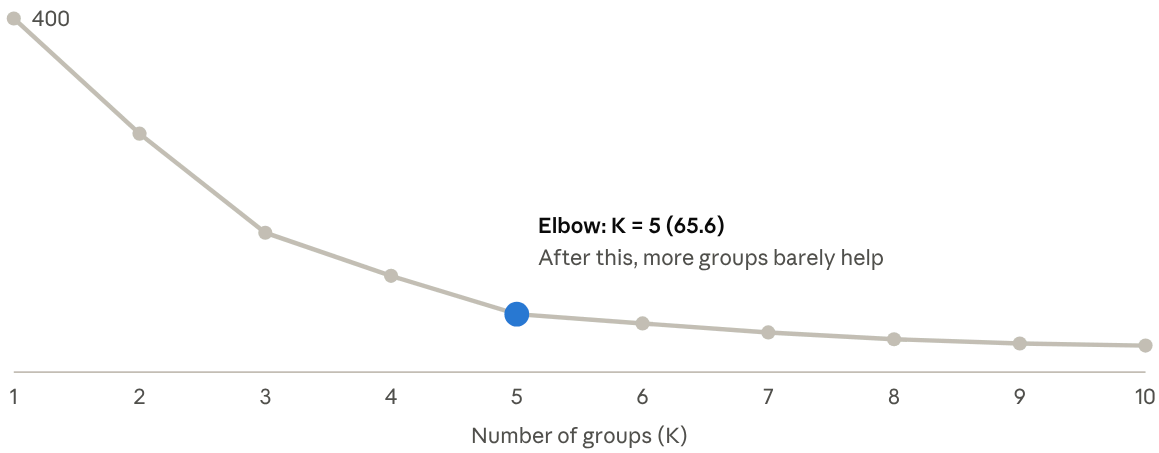
Income and spending were scaled with StandardScaler so both count equally.

Choosing K: the Elbow Method

K-Means was run for K = 1 to 10, measuring inertia each time.

The curve bends at K = 5, so 5 groups is the best choice

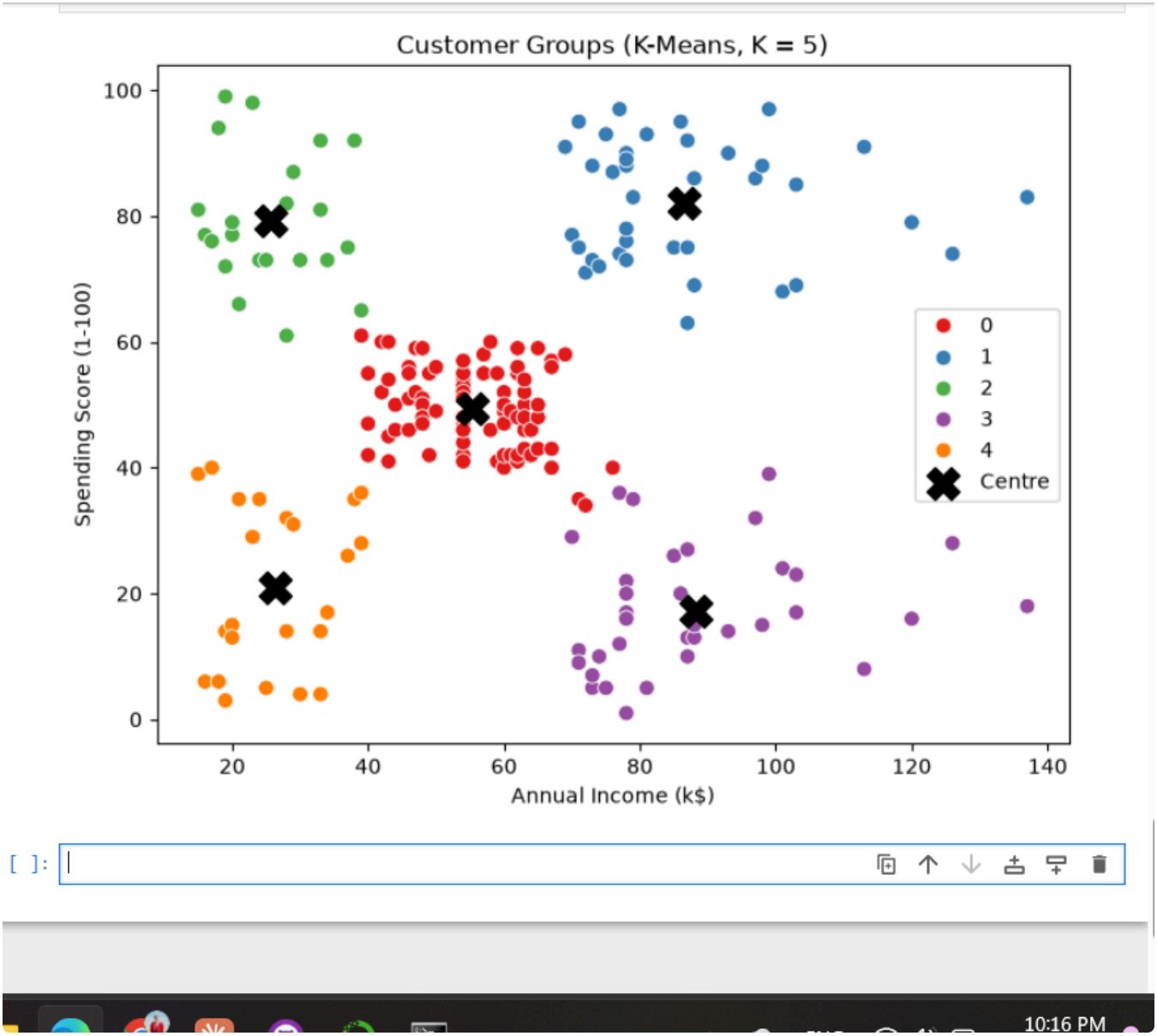
Inertia (how spread out the groups are) for K = 1 to 10; lower is tighter



customer_segmentation.ipynb · K-Means, n_init 10, random_state 42

Inertia falls fast up to K = 5, then flattens, so 5 groups gives tight groups without splitting customers for no reason.

The 5 groups



Each colour is one group and each black X is its centre. The groups sit in the middle and the four corners: low or high income crossed with low or high spending.

Group profiles

Each group's typical customer, from the cluster averages:

Cluster	Name	Customers	Avg income (k$)	Avg spending (1-100)	Avg age	What the mall should do
1	VIP / Big Spenders	39	86.5	82.1	32.7	VIP offers, premium products, early access
0	Average Customers	81	55.3	49.5	42.7	Regular promotions and loyalty points
2	Young Impulse Buyers	22	25.7	79.4	25.3	Discounts and trendy, affordable items
3	Careful Rich	35	88.2	17.1	41.1	Win them over with quality and exclusive deals
4	Budget Savers	23	26.3	20.9	45.2	Budget deals and essentials

Conclusion and next steps

Income and spending alone separate the customers into 5 groups that each need a different marketing approach.

- Add Age as a third feature and see whether the groups change.
- Check group quality with a silhouette score as a second way to choose K.
- Try another clustering method, such as hierarchical clustering, and compare.